Date	**3 JULY 2026**
Team ID	
Project Name	**Credit Card Approval Prediction**
Maximum Marks	**4 Marks**

Project Demonstration Plan

Introduction

The project demonstration was carefully planned to present the complete development process of the **Credit Card Approval Prediction using Machine Learning** system in a clear, organized, and systematic manner. The demonstration was designed to help the audience understand the motivation behind the project, the implementation methodology, the technologies used, and the overall functionality of the developed application. Each stage of the project was explained sequentially, ensuring a smooth flow from problem identification to the final prediction results.

Demonstration Objectives

The primary objective of the demonstration was to showcase how machine learning techniques can be applied to detect fraudulent credit card transactions accurately and efficiently. The demonstration also aimed to explain the complete software development lifecycle followed during the project, including data collection, preprocessing, model development, application deployment, testing, and future enhancement possibilities.

Demonstration Schedule

Step 1: Project Introduction

The demonstration begins with an introduction to the project, explaining the importance of fraud detection in modern financial systems and the challenges associated with manual verification of credit card transactions.

Step 2: Problem Statement

The identified problem is discussed, highlighting the need for an automated system capable of assisting financial institutions in identifying fraudulent transactions while improving accuracy and reducing processing time.

Step 3: Dataset Explanation

The credit card transaction dataset is introduced. The audience is informed about the available features such as **Time**, **Amount**, and anonymized variables **V1 to V28**, along with the target variable representing genuine and fraudulent transactions.

Step 4: Data Preprocessing

The preprocessing phase is explained, including duplicate removal, handling missing values, feature scaling for the **Time** and **Amount** attributes, and preparation of the dataset for machine learning model training.

Step 5: Machine Learning Model Development

The demonstration continues by explaining the training of multiple machine learning models, namely Logistic Regression, Decision Tree, and Random Forest. The evaluation process using Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics is also discussed to justify the selection of the Random Forest classifier.

Step 6: Web Application Demonstration

The Flask-based web application is demonstrated by entering transaction details through the HTML/CSS interface. The complete prediction workflow, including input validation, preprocessing, model loading, prediction generation, and result display, is explained step by step.

Step 7: Testing and Validation

The functional and performance testing performed on the application are presented. The audience is shown how the application handles user input correctly and generates reliable prediction results within a short response time.

Step 8: Future Scope

The demonstration concludes by discussing possible future enhancements, including cloud deployment, database integration, REST API development, explainable AI techniques, mobile application support, and real-time fraud monitoring.

Expected Outcome

By following this demonstration plan, the audience gains a complete understanding of the project objectives, development process, machine learning implementation, application functionality, and practical significance. The demonstration effectively communicates how the proposed solution contributes to improving fraud detection and supports faster, more accurate decision-making in financial institutions.

Conclusion

The planned demonstration provides a structured and comprehensive presentation of the Credit Card Approval Prediction system. It ensures that all important aspects of the project are covered in a logical sequence, enabling the audience to understand both the technical implementation and the real-world benefits of the proposed solution. The demonstration also highlights the scalability of the application and its potential for future improvements.